respect to x_T would instead contribute $\pi_{TS}\varepsilon_{TS}$ to the teacher dynamics. The teacher-side pathway therefore reverses the conventional sign. After each integration step, teacher activity is normalized to unit norm to prevent the linear dynamics from producing unbounded activity. This operation idealizes fast population gain control: in biological circuits, divisive normalization mediated by recurrent and feedforward inhibition constrains the overall response magnitude while largely preserving the relative activity pattern that carries the represented content [**?**].

The student side retains the conventional predictive-coding dynamics. Student activity minimizes its recurrent free energy and receives an excitatory correction from the interface-error population,

$$\tau_S \dot{x}_S = \quad \pi_S M_S^\top \varepsilon_S + \pi_{TS}\varepsilon_{TS}, \tag{7}$$

the student activity is not normalized during the simulation of the whole model.

Both contributions to Eq. (**??**) are gradient-descent terms on the student free energy functional which can therefore be expressed as

$$F_S(x_S, x_T; W_S) = \underbrace{\tfrac{\pi_S}{2}\mathsf{k}M_S x_S\mathsf{k}^2}_{F_S^{\text{rec}}} + \underbrace{\tfrac{\pi_{TS}}{2}\mathsf{k}x_T \quad x_S\mathsf{k}^2}_{F_{TS}}, \qquad \tau_S\dot{x}_S = \ \mathsf{r} \quad _{x_S}F_S, \tag{8}$$

where $\pi_{TS} > 0$ is the precision of the interface prediction error. The student weights follow the local plasticity rule

$$\dot{W}_S = \eta\pi_S\varepsilon_S x_S^\top = \quad \eta\mathsf{r} \ _{W_S}F_S. \tag{9}$$

Activity and plasticity evolve concurrently in the simulations.

The overall influence between S and T deserve carefull reading. Although $\kappa > 0$ denotes an excitatory connection from the interface-error population to the teacher, the student enters $\varepsilon_{TS} = x_T \quad x_S$ with a negative sign, inhibiting the interface error. The combined pathway x_S ! ε_{TS} ! x_T is thus inhibitory: increasing student activity reduces the excitatory error drive reaching the teacher. This effective top-down inhibition is the model counterpart of prefrontal suppression of hippocampal activity. In a conventional predictive-coding network, the interface error would instead inhibit the lower population., producing top down excitation through disinhibition.

[small paragraph with place for citation about why it is bio-plausible to normalize teacher activity]